

M. ZAKI RAMDHAN

Mechanical Engineer · Equipment Reliability & Safety Engineering

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SUMMARY

Mechanical Engineering graduate from Universitas Pertamina (GPA 3.88/4.00, Cum Laude) working where equipment reliability meets safety engineering. During an internship at PT Pertamina Drilling Services Indonesia I ran an FMEA risk assessment on a Blowout Preventer, the critical safety barrier in drilling, and reviewed maintenance history to find recurring failures. I led a funded team developing a safety helmet for the mining industry, designed a machine whose drive is interlocked on two independent sensors, and published first-author research on human error probability. My thesis turns slow hazard simulations into instant hazard zone estimates for emergency response, with a clear statement of where the model stops being valid. Certified Data Analyst under BNSP, bringing Python and data analysis to engineering and safety problems. Looking to apply this in oil and gas, mining, energy or manufacturing, and **open to placement anywhere in Indonesia, including site-based and roster assignments.**

CORE COMPETENCIES

Mechanical and Reliability: FMEA and Risk Priority Number · failure mode analysis · maintenance history review · non-destructive testing · tensile, torsion and impact testing · pump performance testing

Safety and Risk: risk assessment · human reliability assessment (HEART, SLIM) · consequence and dispersion modelling (ALOHA) · root cause analysis (5 Why, Fishbone) · interlock-based safety by design

Engineering Software: SolidWorks · AutoCAD · ANSYS

Data Analysis: Python (pandas, scikit-learn, TensorFlow) · Monte Carlo simulation · factorial experiment design · Excel

EXPERIENCE

PT Pertamina Drilling Services Indonesia (PDSI) · Aug to Sep 2025

Mechanical Engineering Intern, Workshop & Maintenance Division

- Conducted a **risk assessment of the Blowout Preventer (BOP) system**, the most critical safety barrier in drilling operations, using **FMEA**. Calculated Risk Priority Numbers for each failure mode to set maintenance and mitigation priorities.
- Analysed historical maintenance records for drilling equipment to identify recurring failure patterns.
- Performed **non-destructive testing** on drilling equipment components in a heavy equipment workshop.

Universitas Pertamina · 2023 to 2026

Laboratory Teaching Assistant (*three years*)

- **Enforced occupational health and safety protocols** during pressurised hydraulic experiments and destructive testing, supervising 30+ students per semester on high-risk equipment without incident.

- Labs covered: Fluid Mechanics (head loss in piping, pump performance), Mechanics of Materials (tensile, torsion and impact testing on Universal Testing Machines) and Basic Physics.

ENGINEERING & SAFETY PROJECTS

HERO Helmets · Bio-Composite Safety Helmet for the Mining Industry · 2025

Team Leader · Funded by Kemdikbudristek under the PKM-KC scheme

- Led the team from funding proposal through design, fabrication and testing of a safety helmet prototype for mining, made from water hyacinth fibre bio-composite treated with NaOH.
- Ran tensile and impact tests across fibre volume fractions of 0, 5 and 10 percent. The revised treatment lifted the 5 percent specimen by **76.13 percent to 21.03 MPa**, still below the 31.93 MPa unreinforced resin baseline, so the material was reported as promising rather than proven.

Hazard Zone Prediction for Fuel Vapour Dispersion in Tank Overfill Scenarios · 2025 to 2026

Undergraduate Thesis · Supervisors: *Adhitya Ryan Ramadhani, S.T., M.Sc., Ph.D. and Waskito Pranowo, M.T.*

- **The problem:** ALOHA must be run scenario by scenario, which is too slow during an emergency, when a response is needed in seconds.
- Generated **1,215 ALOHA scenarios** through full-factorial design across six atmospheric and operational variables, then built surrogate models predicting **Red, Orange and Yellow zone radii at once**. Red Zone error fell to **1.82 m** against **22.25 m** for the linear baseline, a twelvefold improvement on 243 test scenarios.
- **Mapped where the model stops being valid** through a one-at-a-time robustness study. Inputs outside the training combination space **must be re-verified through ALOHA** before informing any evacuation decision.

Autonomous Trash Bin · Line Follower Waste Collection Robot · Mar to Jul 2025

Project Manager, three-person team · Mechanical Engineering Capstone, Universitas Pertamina

- Owned project coordination, the schedule and Gantt chart, and the frame and structure design. Structural analysis of the aluminium frame gave a **maximum design load of 173.1 kg** at a peak bending moment of **46.3 Nm**.
- **Designed safety into the machine:** the drive is interlocked on two independent measurements, ultrasonic fill height and load cell weight, which must both agree before it moves, so one failed sensor cannot start it. It also halts on an obstructed path until the path clears.

MWT Inspection Reporting Bot · AI Field Safety Walkthrough Reporting · MVP · 2025

- Built an AI bot, designed for **Management Walkthrough (MWT)** reporting in Pertamina field operations, that turns findings sent as free text, photos or voice notes into a structured follow-up record.
- Extracts the inspector, finding, recommendation, PIC, target date and status, runs a **stateful clarification dialog** of up to three questions when information is missing, and archives photo evidence to Drive, **removing manual transcription**.

AntiDeadline.ai · HSSE Root Cause Analysis Platform · Closed beta · 2025 to present

- Turns HSSE and K3 documents into **root cause analysis reports** (5 Why, Fishbone, full RCA) that separate verified facts from assumptions, as drafts for authorised personnel to validate.

LEADERSHIP & ORGANISATION

- **Head of Student Affairs Department**, Mechanical Engineering Student Association (HMTM), Oct 2024 to Jun 2025. Reporting directly to the association chair, I led the department's mentoring, community service and student entrepreneurship programmes.

EDUCATION

Universitas Pertamina, South Jakarta

Bachelor of Mechanical Engineering, Faculty of Industrial Technology · 2022 to 2026

- **GPA 3.88 / 4.00** across 144 credits, **Cum Laude**. SUN UP Scholarship recipient.
- **Graduated 20 August 2026**. Certificate of Graduation issued and valid in place of the diploma while the diploma is being processed.
- Relevant coursework: **Occupational Health and Safety (K3)**, **Risk Management and Engineering**, Rotating Equipment, Maintenance Management, Fluid Mechanics.

PUBLICATIONS

Promoting Evidence-Based Energy Transition: Utilizing Statistical Analysis for Energy and Sustainability Policies in Indonesia. First author. *Sustinere: Journal of Environment and Sustainability* 10(2), 160-175, 2026 (Scopus, SINTA S2). doi.org/10.22515/8h4rm359.

A Comparative Study of Human Reliability Assessment using Success Likelihood Index Method (SLIM) and Human Error Assessment & Reduction Technique (HEART): A case study from a Boeing 737 Max Accident. First author. *Motivection* 7(1), 61-74, 2025. doi.org/10.46574/motivection.v7i1.431.

A Machine-Learning Surrogate for Rapid Hazard-Zone Prediction of n-Hexane Vapor Dispersion from Tank Overfill: Accuracy and Robustness Limits Beyond the Training Domain. First author. Preprint, SSRN, 2026; submitted to *Journal of Loss Prevention in the Process Industries*. doi.org/10.2139/ssrn.7455051.

Studi Kelayakan: Halide Perovskite sebagai Penanda Kimia dalam Industri Minyak dan Gas. Co-author. Universitas Pertamina Press, 2026.

CERTIFICATIONS

- **Certified Data Analyst**, BNSP (Indonesia's national professional certification authority), May 2026, valid for three years.
- **AI Engineering Bootcamp**, Ruangguru Engineering Academy with Skill Academy, 2026. Certificate of Excellence, Grade A (95/100).

ACHIEVEMENTS

- **Gold Medal in Physics**, National Science Olympiad (OSN) 2024.
- **Silver Medal in Physics** and **Bronze Medal in Mathematics**, National Science Week Olympiad 2023.
- **National delegate, ON-MIPA 2023** in Physics.